[2nd Semester] [2025 - 2026]

[Lab 4]
Instructor
Tiba Zaki Abdulhameed, Ph.D.
Lecturer Farah Saad
Lecturer Assistant Ruaa Abdullah

[March 8, 2026]
Email
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Hours
[2 Hours]

Lab 2 – Testing ASP.NET Core APIs using Swagger and Postman

Learning Objectives

By the end of this lab, students will be able to:
- Understand why APIs must be tested without a frontend
- Use Swagger UI to test API endpoints
- Send GET and POST requests using Swagger
- Use Postman to send HTTP requests
- Inspect JSON responses returned from an API

Part 1 – Enabling Swagger

Modify Program.cs to enable Swagger support.

```
var builder = WebApplication.CreateBuilder(args);

// Enable OpenAPI / Swagger
builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();

var app = builder.Build();

if (app.Environment.IsDevelopment())
{
   app.UseSwagger();
   app.UseSwaggerUI();
}

app.MapGet("/greet", (string name) =>
{
   return new
```

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```
    {
        English = $"Hello {name}!",
        Arabic = $"مرحبا {name}!"
    };
});

app.MapPost("/greetpost", (NameRequest request) =>
{
    return new
    {
        English = $"Hello {request.Name}!",
        Arabic = $"مرحبا {request.Name}!"
    };
});

app.Run();

public class NameRequest
{
    public string Name { get; set; }
}
```

Part 2 – Running Swagger

Run the application and open the following URL in your browser:
https://localhost:xxxx/swagger

You will see a list of API endpoints that can be tested directly.

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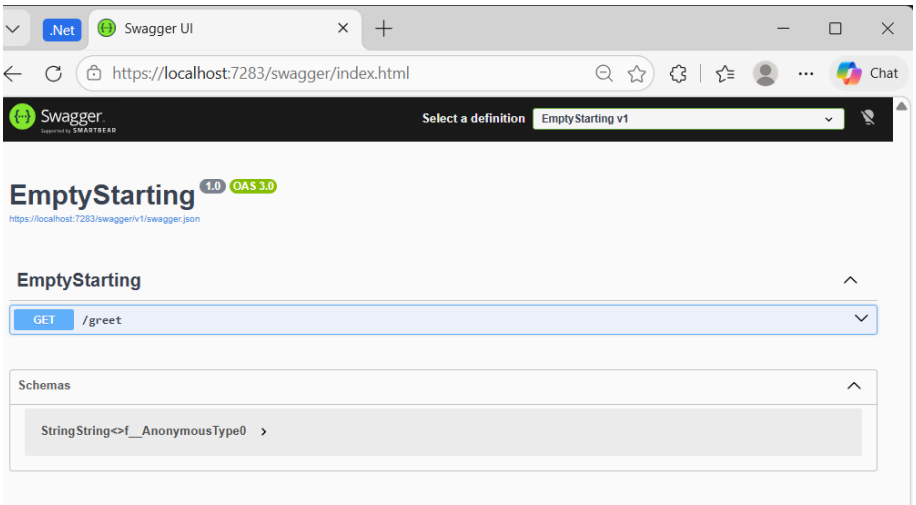
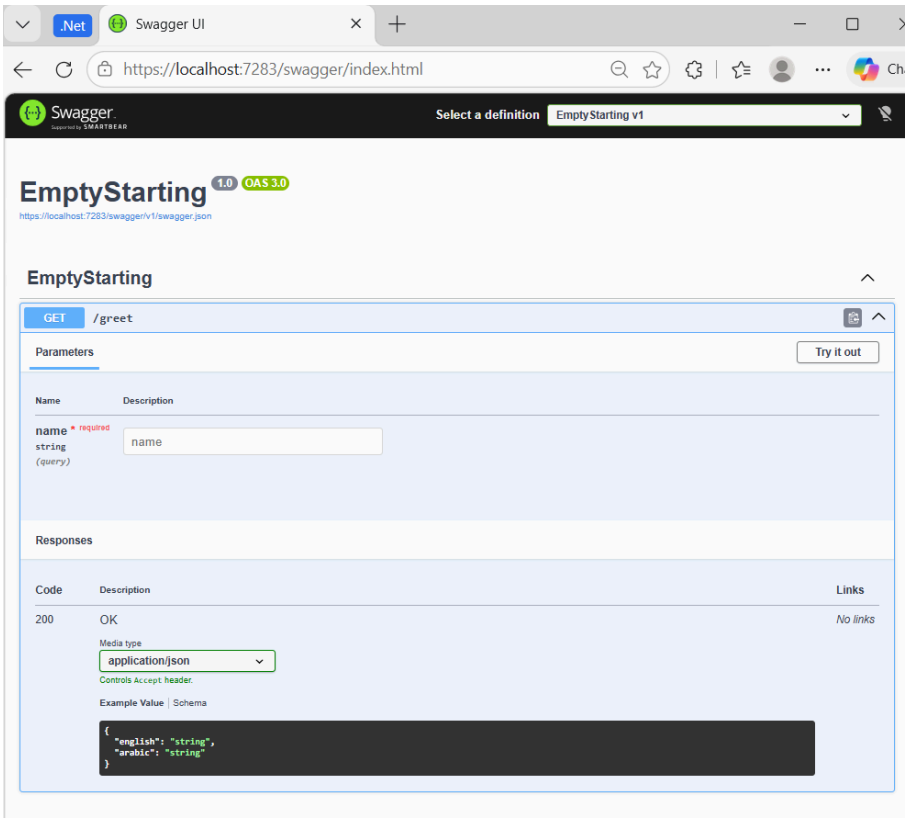
Tiba Zaki Abdulhameed, Ph.D. Lecturer Farah Saad Lecturer Assistant Ruaa Abdullah	Tiba.zaki@nahrainuniv.edu.iq	[2 Hours]





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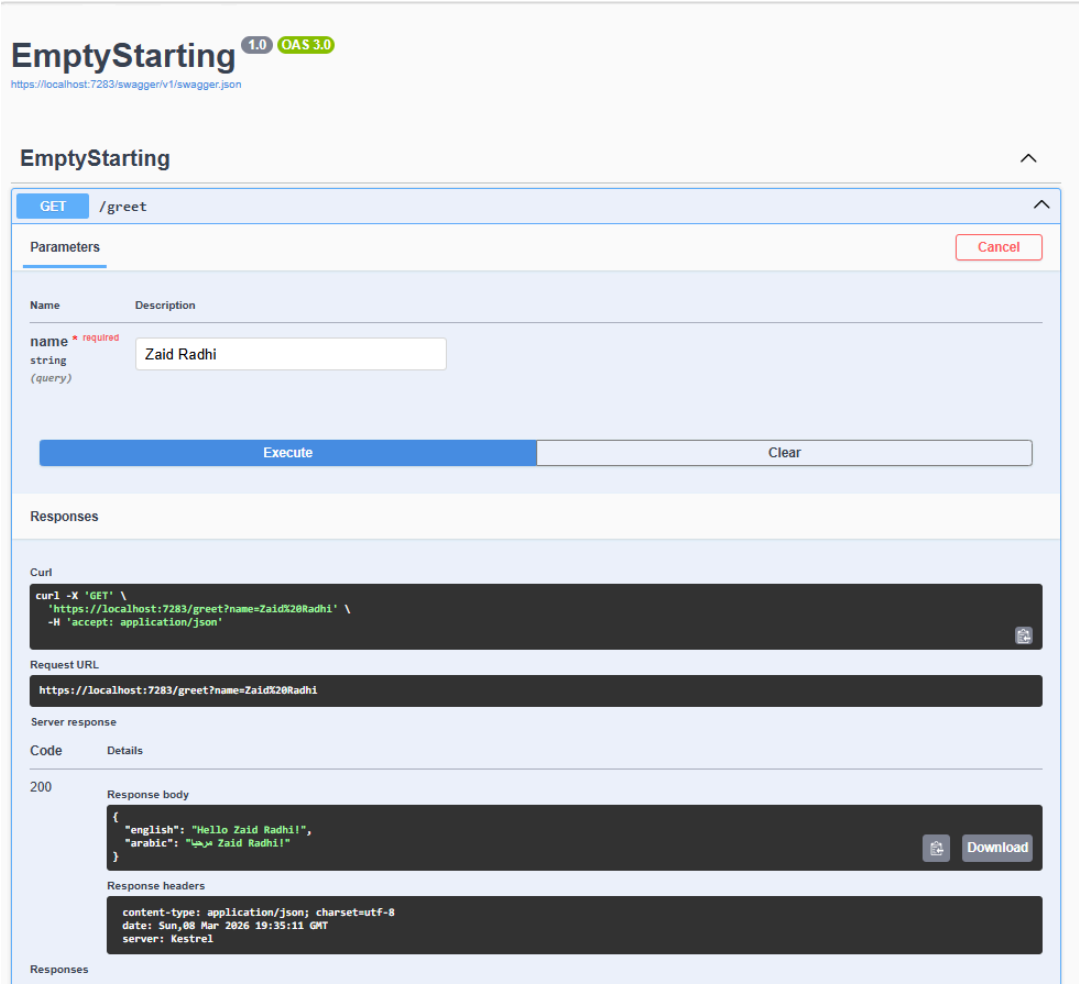

Testing the GET Endpoint

1. Expand GET /greet
2. Click 'Try it out'
3. Enter a name
4. Click Execute

Testing the POST Endpoint

1. Expand POST /greetpost
2. Click 'Try it out'
3. Enter JSON body:

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{ "name": "Ali" }
4. Click Execute

Part 3 – Testing with Postman

Postman is a professional API testing tool widely used in industry.

GET Request in Postman

Method: GET
URL: https://localhost:xxxx/greet?name=Ali
Click Send to view the response.

POST Request in Postman

Method: POST
URL: https://localhost:xxxx/greetpost
Body → raw → JSON:
{ "name": "Ali" }

Student Task

Modify the API so that the response also returns the length of the name.
Example:
{ "english": "Hello Ali!", "arabic": "مرحبا Ali!", "nameLength": 3 }